ADVANCED ANALYSIS OF ALGORITHMS

CS5031 — Week 1: Lectures 1–2

Chapter 1: Introduction to Algorithmic Problems

1.1 What Is an Algorithm?

An algorithm is a finite sequence of clear instructions for solving a problem.

In simple words:

An algorithm tells us what steps to perform, in what order, to obtain the required answer. 5, 8, 2, 10, 4

A simple algorithm could be:

```
1. Assume the first number is the largest.
2. Compare it with the next number.
3. If the next number is larger, update the largest value.
4. Continue until all numbers have been checked.
5. Return the largest value.
```

The answer is:

```
</> Plain text

10
```

An algorithm is therefore different from a **program**.

An algorithm describes the logical solution, while a program is the implementation of that algorithm in a programming language such as Python, C++, Java, or C.

1.2 What Is Algorithm Analysis?

Two different algorithms may solve exactly the same problem but require very different amounts of time.

Therefore, in **Design and Analysis of Algorithms**, we normally ask two major questions:

Design question

How can we solve the problem?

Analysis question

How efficient is our solution?

Efficiency usually includes:

- **time efficiency** — how long the algorithm takes;
- **space efficiency** — how much memory the algorithm requires.

The central lesson of these slides is:

> Choosing the correct algorithm can make an enormous difference.

This becomes particularly clear when we compare two different algorithms for computing Fibonacci numbers later in this chapter.

1.3 Types of Problems

The slides divide problems into three broad categories:

1. straightforward/simple programming problems;
2. algorithmic problems;
3. artificial intelligence problems.

The course primarily focuses on problems that are clearly formulated but difficult to solve efficiently.

Week1-merged

1.4 Straightforward Programming Problems

A **straightforward programming problem** is one for which the natural solution is already reasonably efficient.

The slides describe these problems as having a straightforward implementation and an already efficient natural solution. Week1-merged

Examples shown in the slides include:

Example 1: Display Given Text

Suppose the computer receives:

```
Plain text

Design and Analysis of Algorithms
```

The task is simply to display the same text.

There is no complicated algorithmic decision involved.

Example 2: Copy a File

Suppose we want to copy:

```
Plain text

source.txt
```

to:

```
backup.txt
```

Conceptually, the computer:

```
reads data from source.txt
writes the data to backup.txt
```

Every piece of information must normally be read and written.

There is no magical algorithm that avoids processing the data itself.

1.5 Simple Programming Problems

Another example in the slides is searching for a word in a piece of text.

Suppose we have:

```
The design and analysis of algorithms is important.
```

and want to find:

```
analysis
```

One natural technique is a **linear scan**.

The slides specifically describe simple programming problems as ones where a linear scan may be necessary and where the obvious approach already works reasonably well. Week1-merged

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1.6 Linear Scan

A **linear scan**, also called **linear search**, means examining elements one after another.

Suppose:

```
A = [13, 7, 21, 5, 18]
```

and we want to find:

```
5
```

The search proceeds as follows:

Position	Value	Is it 5?
0	13	No
1	7	No
2	21	No
3	5	Yes

The algorithm stops after finding 5.

A basic version is:

```
LinearSearch(A, target)

for each element x in A:
    if x = target:
        return FOUND

return NOT_FOUND
```

If there are n elements, the algorithm may have to examine all n elements.

Therefore, its running time is:

$$O(n)$$

The notation $O(n)$ will be studied more formally later, but for now it simply means:

> As the number of elements grows, the amount of work grows approximately in direct proportion to the number of elements.

1.7 Algorithmic Problems

Algorithmic problems are more interesting.

The slides describe them as problems where:

- it may not initially be clear what to do;
- simple ideas may be too slow;
- there is room for optimization. Week1-merged

This is where **algorithm design** becomes important.

1.8 Example: Shortest Path Problem

The slides give the example:

> Find the shortest path between locations.

Imagine the following road network:

```
A ----4---- B
|           |
2           3
|           |
C ----1---- D
```

The numbers could represent distances.

Suppose we want to travel from A to D.

Possible paths include:

```
A → B → D
distance = 4 + 3 = 7
```

and

```
```

```
A → C → D
distance = 2 + 1 = 3
```

Therefore:

```
Plain text
A → C → D
```

is the shorter route.

Important terminology

A road-network problem is often represented using a **graph**.

A graph contains:

Vertex or node:
An object or location.

For example:

```
Plain text
A, B, C, D
```

Edge:
A connection between two vertices.

Weight:
A numerical value associated with an edge, such as:

- distance;
- time;
- cost;
- fuel consumption.

The course slides introduce the problem rather than a particular shortest-path algorithm.

As additional background, common algorithms for shortest paths include **Breadth-First Search** for some unweighted graphs and **Dijkstra's algorithm** for many non-negative weighted graphs.

1.9 Example: Assigning Students to Dorm Rooms

Another slide asks us to:

> Find the best assignment of students to dorm rooms.

Suppose:

```
Plain text

Student A prefers Room 2
Student B prefers Room 1
Student C prefers Room 3
```

The problem is not simply assigning any room.

We may want an assignment that maximizes overall satisfaction.

This belongs to a family of problems known as **assignment problems**.

An assignment problem normally contains:

- a set of objects;
- another set of resources;
- preferences or costs;
- a goal such as minimizing cost or maximizing satisfaction.

Additional information

These problems can often be represented using:

- bipartite graphs;
- matching;
- optimization techniques.

The **Hungarian algorithm** is one well-known algorithm for the classical assignment problem, although it is not introduced in these Week 1 slides.

1.10 Example: Measuring Similarity Between Documents

The slides also give:

> Measure similarity of documents.

Suppose we have:

Document A

```
algorithms improve computer performance
```

Document B

```
efficient algorithms improve performance
```

These documents contain several related words.

A computer may calculate a numerical **similarity score**.

For example:

```
0      → completely different
1      → extremely similar
```

Additional background

Common ways of comparing documents include:

- word overlap;
- Jaccard similarity;
- cosine similarity;
- TF-IDF vectors;
- modern neural embeddings.

The important idea for this course is that it is not always immediately obvious which computational procedure gives the best answer efficiently.

1.11 Optimization

The term **optimization** means improving a solution according to some objective.

For example:

```
Find the shortest route.
```

The objective is:

```
minimize distance
```

In student-room assignment:

```
maximize satisfaction
```

In scheduling:

```
minimize completion time
```

Optimization does **not** simply mean making code shorter.

It normally means finding the best or a better solution according to a measurable criterion.

1.12 Artificial Intelligence Problems

The slides distinguish some AI problems because even clearly defining the desired result can sometimes be difficult.

Examples in the slides include:

- understanding natural language;
- identifying objects in photographs;
- playing games well.

These are shown as examples that go beyond the primary scope of this introductory algorithms material. Week1-merged

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Natural Language Understanding

Consider:

```
I saw the man with the telescope.
```

This sentence can mean:

1. I used a telescope to see the man.

or:

2. I saw a man who was carrying a telescope.

Humans use context to understand language.

For a machine, this can be difficult.

Object Recognition

Given an image, a computer may need to answer:

```
Plain text

Is this a cat?
Is this a car?
Where is the person?
```

This requires interpretation rather than simple arithmetic.

Game Playing

In chess, for example, the computer must decide:

```
Plain text

Which move should I make?
```

There may be an enormous number of possible future move sequences.

Modern AI still relies heavily on algorithms, but these slides separate such problems because their formulation and evaluation can be more complex than the classical algorithmic problems being introduced.

Chapter 2: Fibonacci Numbers

The next major topic in the slides is the **Fibonacci sequence**.

The learning objective is not merely to calculate Fibonacci numbers. The real goal is to demonstrate how a seemingly reasonable algorithm can be extremely slow, while a better-designed algorithm can

solve the same problem efficiently. Week1-merged

2.1 Definition of Fibonacci Numbers

The Fibonacci sequence is defined as:

$$F_n = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ F_{n-1} + F_{n-2} & n > 1 \end{cases}$$

The slides therefore generate:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

Week1-merged

2.2 Understanding the Notation

F_n

Means:

> the Fibonacci number at position n.

For example:

$$F_0 = 0$$

$$F_1 = 1$$

$$F_2 = 1$$

$$F_3 = 2$$

F_{n-1}

Means:

> the previous Fibonacci number.

F_{n-2}

Means:

the Fibonacci number two positions earlier.

2.3 Calculating Fibonacci Numbers Manually

Starting with:

$$F_0 = 0$$

and

$$F_1 = 1$$

we obtain:

$$F_2 = F_1 + F_0$$

$$= 1 + 0 = 1$$

Then:

$$F_3 = F_2 + F_1$$

$$= 1 + 1 = 2$$

Then:

$$F_4 = F_3 + F_2$$

$$= 2 + 1 = 3$$

Then:

$$F_5 = F_4 + F_3$$

$$= 3 + 2 = 5$$

Then:

$$F_6 = F_5 + F_4$$

$$= 5 + 3 = 8$$

So:

Plain text

```
F0 = 0
F1 = 1
F2 = 1
F3 = 2
F4 = 3
F5 = 5
F6 = 8
```

2.4 Recurrence Relation

The expression

$$F_n = F_{n-1} + F_{n-2}$$

is called a **recurrence relation**.

A recurrence relation defines a value using previous values of the same sequence.

For example:

$$F_8$$

depends on:

$$F_7$$

and:

$$F_6$$

This recursive mathematical definition naturally suggests a recursive computer algorithm.

As we will see, however:

> A mathematical definition that looks elegant does not automatically produce an efficient program.

2.5 Base Cases

The values

$$F_0 = 0$$

and

$$F_1 = 1$$

are called **base cases**.

A base case is a situation whose answer is known directly.

Base cases are extremely important in recursion because they stop the recursive calls.

Without an appropriate base case, recursion may continue indefinitely.

2.6 Historical Motivation: Rabbit Population

The slides mention that Fibonacci numbers were developed in connection with a rabbit population problem.

The classical simplified model assumes things such as:

- rabbit pairs mature;
- mature pairs reproduce;
- rabbits do not die during the experiment;
- reproduction happens at regular intervals.

Under these idealized assumptions, the number of rabbit pairs follows a Fibonacci-like pattern.

The important point is not biological realism. It is that a recurrence relation can model how one state develops from previous states.

2.7 Fibonacci Numbers Grow Rapidly

The slides provide the following lemma:

$$F_n \geq 2^{n/2}$$

for:

$$n \geq 6$$

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This inequality tells us that Fibonacci numbers grow at least exponentially fast.

2.8 What Is a Lemma?

A **lemma** is a mathematical statement that has been proved and is often used to prove another important result.

A lemma is not simply a guess.

It is a proven result.

You may think of:

```
Plain text

Theorem → major mathematical result
Lemma   → supporting proven result
```

The distinction is mostly about purpose and importance rather than a difference in mathematical validity.

2.9 Proof by Induction

The slides prove the rapid-growth result using **mathematical induction**.

Induction normally contains:

1. a base case;
2. an induction hypothesis;
3. an inductive step.

Base Cases

For $n = 6$:

$$F_6 = 8$$

and:

$$2^{6/2} = 2^3 = 8$$

Therefore:

$$F_6 \geq 2^{6/2}$$

is true.

For $n = 7$:

$$F_7 = 13$$

while:

$$2^{7/2} \approx 11.31$$

Therefore:

$$13 \geq 11.31$$

is also true.

2.10 Induction Hypothesis

Assume that the result is true for the previous required Fibonacci numbers:

$$F_{n-1} \geq 2^{(n-1)/2}$$

and:

$$F_{n-2} \geq 2^{(n-2)/2}$$

Because Fibonacci uses two previous values, this proof effectively uses the idea of **strong induction**.

2.11 Inductive Step

We know:

$$F_n = F_{n-1} + F_{n-2}$$

Using the induction assumptions:

$$F_n \geq 2^{(n-1)/2} + 2^{(n-2)/2}$$

Since:

$$2^{(n-1)/2} \geq 2^{(n-2)/2}$$

we have:

$$F_n \geq 2\left(2^{(n-2)/2}\right)$$

Using:

$$2 = 2^1$$

we get:

$$F_n \geq 2^{1+(n-2)/2}$$

and:

$$1 + \frac{n-2}{2} = \frac{n}{2}$$

therefore:

$$\boxed{F_n \geq 2^{n/2}}$$

The result has been proved.

2.12 Closed-Form Formula for Fibonacci Numbers

The slides also provide the direct formula:

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$$

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This formula is commonly known as **Binet's Formula**.

2.13 The Golden Ratio

Define:

$$\phi = \frac{1+\sqrt{5}}{2}$$

Approximately:

$$\phi \approx 1.6180339887$$

This is called the **golden ratio**.

The second quantity is:

$$\psi = \frac{1-\sqrt{5}}{2}$$

Approximately:

$$\psi \approx -0.618$$

So Binet's formula becomes:

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$$

Since:

$$|\psi| < 1$$

the value:

$$\psi^n$$

becomes extremely small as n becomes large.

Therefore approximately:

$$F_n \approx \frac{\phi^n}{\sqrt{5}}$$

This explains why Fibonacci numbers grow exponentially.

2.14 Large Fibonacci Numbers

The slides give examples including:

$$F_{20} = 6765$$

$$F_{50} = 12586269025$$

$$F_{100} = 354224848179261915075$$

demonstrating that Fibonacci numbers become very large very quickly. Week1-merged

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Chapter 3: Computing Fibonacci Numbers

The computational problem is:

Input

An integer:

$$n \geq 0$$

Output

$$F_n$$

This input/output formulation is explicitly used in the slides. Week1-merged

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3.1 Input and Output

An **input** is the information supplied to an algorithm.

An **output** is the answer produced by the algorithm.

For example:

```
Plain text

Input:
n = 7

Output:
13
```

because:

$$F_7 = 13$$

3.2 First Solution: Recursive Fibonacci Algorithm

The slides introduce:

```
Plain text

FibRecurs(n)

if n ≤ 1:
    return n
else:
    return FibRecurs(n − 1) + FibRecurs(n − 2)
```

Week1-merged

This algorithm directly follows the mathematical definition.

3.3 Meaning of Recursion

Recursion occurs when a function calls itself.

For example:

```
Plain text

FibRecurs(5)
```

calls:

```
Plain text

FibRecurs(4)
```

and:

```
Plain text

FibRecurs(3)
```

Those calls then make additional calls.

3.4 Understanding the Base Condition

Consider:

```
Plain text

if n ≤ 1:
    return n
```

If:

```
Plain text

n = 0
```

the function returns:

```
Plain text

0
```

If:

```
Plain text

n = 1
```

the function returns:

```
Plain text

1
```

No further recursive call is needed.

3.5 Dry Run of FibRecurs(5)

Let us perform a complete conceptual dry run.

We begin with:

```
Plain text

FibRecurs(5)
```

Since:

```
Plain text

5 > 1
```

we calculate:

```
Plain text

FibRecurs(4) + FibRecurs(3)
```

Now:

```
Plain text

FibRecurs(4)
```

```
= FibRecurs(3) + FibRecurs(2)
```

and:

Plain text

```
FibRecurs(3)
= FibRecurs(2) + FibRecurs(1)
```

Also:

Plain text

```
FibRecurs(2)
= FibRecurs(1) + FibRecurs(0)
= 1 + 0
= 1
```

Therefore:

Plain text

```
FibRecurs(3)
= 1 + 1
= 2
```

Then:

Plain text

```
FibRecurs(4)
= 2 + 1
= 3
```

Finally:

Plain text

```
FibRecurs(5)
= FibRecurs(4) + FibRecurs(3)
= 3 + 2
= 5
```

Therefore:

$$\boxed{F_5 = 5}$$

3.6 Recursion Tree

The calls can be represented as:

Plain text

```
                  Fib(5)
               /          \
           Fib(4)         Fib(3)
          /      \        /     \
      Fib(3)    Fib(2) Fib(2)  Fib(1)
      /   \      /  \    /  \
  Fib(2) Fib(1) F1  F0  F1  F0
   /  \
  F1  F0
```

Look carefully at this tree.

`Fib(3)` is calculated more than once.

`Fib(2)` is calculated several times.

Even `Fib(1)` and `Fib(0)` are repeatedly calculated.

This **repeated computation** is the main reason why the algorithm is inefficient.

3.7 What Is Running Time?

The **running time** of an algorithm measures how much computational work it performs.

The slides define:

$$T(n)$$

as the number of lines of code executed by `FibRecurs(n)` . Week1-merged

This gives us a mathematical way to describe the amount of work.

3.8 Running Time for the Base Case

When:

$$n \leq 1$$

the slides count:

$$T(n) = 2$$

because only a small constant amount of work is performed.

3.9 Running Time for $n \geq 2$

For larger values:

$$T(n) = T(n-1) + T(n-2) + 3$$

The two recursive terms represent:

```
Plain text
FibRecurs(n - 1)
FibRecurs(n - 2)
```

while the constant `3` represents the additional operations performed in the current function invocation. Week1-merged

---

3.10 Small Dry Run of the Running-Time Recurrence

Given:

$$T(0) = 2$$

and:

$$T(1) = 2$$

then:

$$T(2) = 3 + T(1) + T(0)$$

$$= 3 + 2 + 2$$

$$= 7$$

For $n = 3$:

$$T(3) = 3 + T(2) + T(1)$$

$$= 3 + 7 + 2$$

$$= 12$$

For $n = 4$:

$$T(4) = 3 + 12 + 7$$

$$= 22$$

For $n = 5$:

$$T(5) = 3 + 22 + 12$$

$$= 37$$

The work starts increasing very rapidly.

3.11 Why Is Recursive Fibonacci So Slow?

The recurrence has essentially the same branching pattern as Fibonacci itself.

The slides conclude:

$$T(n) \geq F_n$$

and give:

$$T(100) \approx 1.77 \times 10^{21}$$

operations, which they estimate would require roughly **56,000 years at 1 GHz**. Week1-merged

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3.12 Understanding the 1 GHz Example

A processor operating at:

$$1 \text{ GHz}$$

has a clock frequency of approximately:

$$10^9$$

cycles per second.

If we roughly imagine one counted operation per cycle, then:

$$\frac{1.77 \times 10^{21}}{10^9} = 1.77 \times 10^{12}$$

seconds.

Converting to years:

$$\frac{1.77 \times 10^{12}}{60 \times 60 \times 24 \times 365} \approx 56,000$$

years.

This is only an illustrative model—real processors and instructions are more complicated—but it makes the scale of the problem clear.

3.13 Time Complexity of Recursive Fibonacci

A more standard asymptotic analysis gives approximately:

$$O(\phi^n)$$

where:

$$\phi \approx 1.618$$

It is also common to use the looser upper bound:

$$O(2^n)$$

Both tell us the important thing:

> The naive recursive Fibonacci algorithm has exponential running time.

Exponential algorithms become impractical very quickly.

3.14 Space Complexity of Recursive Fibonacci

Even though many recursive calls are created, they are not all simultaneously stored.

The maximum recursion depth is approximately:

$$n$$

Therefore the recursion stack requires:

$$O(n)$$

space.

This is useful additional information when analyzing an algorithm because time is not the only resource algorithms consume.

Chapter 4: Efficient Fibonacci Algorithm

The slides next ask:

> Instead of repeatedly recomputing previous Fibonacci numbers, why not compute each value once and keep it?

This produces a dramatically better algorithm.

4.1 Hand Calculation

Humans calculate:

```
Plain text
0, 1
```

then:

```
Plain text
0 + 1 = 1
```

so:

```
Plain text
0, 1, 1
```

Then:

```
Plain text
1 + 1 = 2
```

giving:

```
0, 1, 1, 2
```

Then:

```
1 + 2 = 3
```

giving:

```
0, 1, 1, 2, 3
```

and so on.

The slides use this observation to motivate the efficient algorithm.

4.2 FibList Algorithm

The algorithm in the slides is:

```
FibList(n)

create an array F[0 ... n]

F[0] ← 0
F[1] ← 1

for i from 2 to n:
    F[i] ← F[i − 1] + F[i − 2]

return F[n]
```

Week1-merged

4.3 What Is an Array?

An **array** is a collection of elements stored using indexed positions.

For example:

```
Plain text

F = [0, 1, 1, 2, 3, 5, 8]
```

The indexes are:

```
Plain text

Index: 0  1  2  3  4  5  6
Value: 0  1  1  2  3  5  8
```

Therefore:

```
Plain text

F[0] = 0
F[1] = 1
F[5] = 5
F[6] = 8
```

4.4 Meaning of the Assignment Arrow

The slides use:

$$\leftarrow$$

For example:

```
Plain text

F[0] ← 0
```

This means:

> Store the value 0 in `F[0]`.

Similarly:

```
Plain text
```

```
F[i] ← F[i − 1] + F[i − 2]
```

means:

> Calculate the sum and store it in `F[i]` .

The assignment symbol should not be confused with mathematical equality.

4.5 Dry Run of FibList(6)

We want:

$$F_6$$

Initially:

```
Plain text

F[0] = 0
F[1] = 1
```

Iteration $i = 2$

```
Plain text

F[2] = F[1] + F[0]
     = 1 + 0
     = 1
```

Array:

```
Plain text

[0, 1, 1]
```

Iteration $i = 3$

```
Plain text

F[3] = F[2] + F[1]
     = 1 + 1
     = 2
```

Array:

```
Plain text

[0, 1, 1, 2]
```

Iteration $i = 4$

```
Plain text

F[4] = F[3] + F[2]
     = 2 + 1
     = 3
```

Array:

```
Plain text

[0, 1, 1, 2, 3]
```

Iteration $i = 5$

```
Plain text

F[5] = F[4] + F[3]
     = 3 + 2
     = 5
```

Array:

```
Plain text

[0, 1, 1, 2, 3, 5]
```

Iteration $i = 6$

```
Plain text

F[6] = F[5] + F[4]
     = 5 + 3
     = 8
```

Final array:

```
Plain text

[0, 1, 1, 2, 3, 5, 8]
```

Return:

```
Plain text

8
```

Therefore:

$$\boxed{F_6 = 8}$$

4.6 Dry-Run Table

i	$F[i-2]$	$F[i-1]$	Calculation	$F[i]$
2	0	1	$0+1$	1
3	1	1	$1+1$	2
4	1	2	$1+2$	3
5	2	3	$2+3$	5
6	3	5	$3+5$	8

4.7 Running Time

The slides count the running time as:

$$T(n) = 2n + 2$$

so:

$$T(100) = 202$$

under their line-counting model. Week1-merged

Compare this with the recursive algorithm's enormous number of operations.

This is an extraordinary improvement.

4.8 Big-O Complexity

Ignoring constants and smaller terms:

$$2n + 2$$

becomes:

$$O(n)$$

Therefore:

```
Plain text

FibList time complexity = O(n)
```

This is called **linear time**.

4.9 Comparison of the Two Fibonacci Algorithms

Feature	Recursive FibRecurs	Iterative FibList
Main idea	Follow recurrence directly	Compute values once
Repeated calculations	Yes	No
Time complexity	Exponential	$O(n)$
Approx. behavior	Very slow	Very fast
Space	$O(n)$ recursion stack	$O(n)$ array
Suitable for large n	No	Much better

The slides summarize the key lesson as:

> The right algorithm makes all the difference. Week1-merged

4.10 Extra Improvement: Fibonacci with Constant Space

The slides store every Fibonacci number in an array.

But if we need only F_n, we actually need only the previous two values.

We can write:

Plain text

```
FibEfficient(n)

if n ≤ 1:
    return n

previous ← 0
current  ← 1

for i from 2 to n:
    next ← previous + current
    previous ← current
    current ← next

return current
```

For example, for $n = 6$:

i	previous	current	next
Start	0	1	—
2	0	1	1
3	1	1	2
4	1	2	3
5	2	3	5
6	3	5	8

This version has:

$$O(n)$$

time but only:

$$O(1)$$

extra space.

Important

This constant-space version is **additional information**. The Week 1 slides use the array-based `FibList` algorithm.

Chapter 5: Greatest Common Divisor — GCD

The final major part of the slides introduces the **Greatest Common Divisor**.

For integers a and b, the slides define:

$$\gcd(a, b)$$

as the largest integer d that divides both a and b. Week1-merged

Usually we take the GCD to be non-negative.

5.1 What Does "Divides" Mean?

The notation:

$$d \mid a$$

is read as:

> d divides a.

It means that there is some integer k such that:

$$a = d \times k$$

For example:

$$3 \mid 12$$

because:

$$12 = 3 \times 4$$

Similarly:

$$4 \mid 20$$

because:

$$20 = 4 \times 5$$

But:

$$5 \nmid 12$$

because 12 cannot be written as 5 multiplied by an integer.

5.2 Example of GCD

Consider:

$$12$$

and:

$$18$$

Divisors of 12:

```
Plain text
1, 2, 3, 4, 6, 12
```

Divisors of 18:

```
Plain text
1, 2, 3, 6, 9, 18
```

Common divisors:

```
Plain text
1, 2, 3, 6
```

The greatest one is:

```
Plain text
6
```

Therefore:

$$\boxed{\gcd(12, 18) = 6}$$

5.3 GCD and Simplifying Fractions

The slides motivate GCD using fraction simplification.

Consider:

$$\frac{18}{24}$$

The GCD is:

$$\gcd(18, 24) = 6$$

Divide numerator and denominator by 6:

$$\frac{18/6}{24/6} = \frac{3}{4}$$

Therefore:

$$\boxed{\frac{18}{24} = \frac{3}{4}}$$

The largest common divisor removes all common factors in one step.

5.4 Applications of GCD

The slides specifically mention connections with:

Number Theory

Number theory is the branch of mathematics concerned with integers and their properties.

Topics include:

- divisibility;
- primes;
- factors;
- GCD;
- modular arithmetic.

Cryptography

Cryptography studies methods for protecting information.

GCD computations occur in important cryptographic mathematics, including modular arithmetic and public-key techniques.

The Euclidean algorithm is particularly useful because it calculates GCDs efficiently even when numbers are extremely large.

5.5 Computational GCD Problem

The problem can be written formally as:

Input

Two integers:

$$a, b \geq 0$$

Output

$$\gcd(a, b)$$

The slides use a large example:

$$\gcd(3918848, 1653264)$$

This is deliberately large enough to show why checking divisors one by one is not attractive.

Chapter 6: Naive GCD Algorithm

The first algorithm in the slides is:

```
NaiveGCD(a, b)

best ← 0

for d from 1 to a + b:
    if d | a and d | b:
        best ← d

return best
```

The slides estimate its running time as approximately $a + b$, which becomes impractical for very large values. Week1-merged

---

6.1 Meaning of `best`

The variable:

```
Plain text

best
```

stores the largest common divisor discovered so far.

Initially:

```
Plain text

best = 0
```

Whenever another common divisor is discovered, it replaces `best` .

6.2 Dry Run: NaiveGCD(18, 12)

Start:

```
Plain text

best = 0
```

According to the slide algorithm:

```
Plain text

d = 1 to 18 + 12
d = 1 to 30
```

Now test each candidate.

d	$d \mid 18$?	$d \mid 12$?	Update `best` ?
1	Yes	Yes	1
2	Yes	Yes	2
3	Yes	Yes	3
4	No	Yes	No
5	No	No	No
6	Yes	Yes	6
7 onward	No larger common divisor	—	remains 6

At the end:

```
Plain text
best = 6
```

Return:

$$\boxed{6}$$

6.3 Why Is the Naive Algorithm Slow?

Suppose:

$$a$$

contains 20 decimal digits.

It may be approximately:

$$10^{20}$$

A loop that performs roughly 10^{20} iterations is completely impractical.

This reveals an important distinction:

> An algorithm can be correct and still be a bad algorithm.

Correctness answers:

```
Plain text

Does it produce the right answer?
```

Efficiency answers:

```
Plain text

How much work is required?
```

A good algorithm usually needs both.

6.4 Additional Note About the Naive Algorithm

We know that a common divisor cannot exceed:

$$\min(a, b)$$

so a slightly better naive algorithm could stop there instead of $a + b$.

However, this does **not** solve the fundamental problem.

For very large numbers, checking every candidate divisor is still extremely slow.

What we need is a completely different mathematical idea.

That idea is the **Euclidean Algorithm**.

Chapter 7: Euclidean Algorithm

The Euclidean Algorithm is one of the oldest known algorithms.

Its power comes from an important mathematical property of GCD.

7.1 Division with Remainder

Suppose we divide integer a by integer b.

We can write:

$$a = bq + r$$

where:

- q is the **quotient**;
- r is the **remainder**.

The remainder satisfies:

$$0 \leq r < b$$

when $b > 0$.

7.2 Example

Divide:

$$17$$

by:

$$5$$

We obtain:

$$17 = 5(3) + 2$$

Therefore:

```
Plain text

quotient  = 3
remainder = 2
```

We can write:

$$17 \bmod 5 = 2$$

The symbol:

```
Plain text

mod
```

means **modulo**, or remainder after division.

7.3 Key Lemma Behind Euclid's Algorithm

The slides state that if a' is the remainder when a is divided by b, then:

$$\gcd(a, b) = \gcd(a', b) = \gcd(b, a')$$

Week1-merged

Using the more usual symbol r:

$$\boxed{\gcd(a, b) = \gcd(b, r)}$$

where:

$$r = a \bmod b$$

This is the central idea of the Euclidean Algorithm.

7.4 Why Is the Lemma True?

Suppose:

$$a = bq + r$$

Rearranging:

$$r = a - bq$$

Suppose d divides both a and b.

Then:

$$a = dx$$

and:

$$b = dy$$

for some integers x and y.

Now:

$$r = a - bq$$

Substitute:

$$r = dx - dyq$$

Factor out d:

$$r = d(x - yq)$$

Therefore d also divides r.

So every common divisor of a and b is also a common divisor of b and r.

The reverse argument also works.

Therefore they have exactly the same common divisors, which means:

$$\boxed{\gcd(a, b) = \gcd(b, r)}$$

This simple mathematical observation produces a dramatically faster algorithm.

7.5 Euclidean Algorithm Pseudocode

The slides give:

Plain text

```
EuclidGCD(a, b)

if b = 0:
    return a

a' ← remainder when a is divided by b

return EuclidGCD(b, a')
```

Week1-merged

A programming-style equivalent is:

Plain text

```
EuclidGCD(a, b)

if b == 0:
    return a

return EuclidGCD(b, a % b)
```

7.6 Why Does `b = 0` Stop the Algorithm?

For any positive integer a:

$$\gcd(a, 0) = a$$

Why?

Every positive divisor of a also divides zero because:

$$0 = a \times 0$$

The greatest divisor shared by a and zero is therefore a itself.

So when the second value reaches zero, the first value is the answer.

7.7 Small Dry Run: gcd(18, 12)

Start:

$$\gcd(18, 12)$$

Compute:

$$18 \bmod 12 = 6$$

Therefore:

$$\gcd(18, 12) = \gcd(12, 6)$$

Next:

$$12 \bmod 6 = 0$$

Therefore:

$$\gcd(12, 6) = \gcd(6, 0)$$

Now the second argument is zero.

Return:

$$\boxed{6}$$

Only two remainder operations were required.

Compare this with checking candidate divisors one by one.

7.8 Full Dry Run from the Slides

The slides use:

$$\gcd(3918848, 1653264)$$

Let us perform the complete calculation.

Step 1

$$3918848 = 1653264(2) + 612320$$

Therefore:

$$\gcd(3918848, 1653264) = \gcd(1653264, 612320)$$

Step 2

$$1653264 = 612320(2) + 428624$$

Therefore:

$$\gcd(1653264, 612320) = \gcd(612320, 428624)$$

Step 3

$$612320 = 428624(1) + 183696$$

Therefore:

$$\gcd(612320, 428624) = \gcd(428624, 183696)$$

Step 4

$$428624 = 183696(2) + 61232$$

Therefore:

$$\gcd(428624, 183696) = \gcd(183696, 61232)$$

Step 5

$$183696 = 61232(3) + 0$$

Therefore:

$$\gcd(183696, 61232) = \gcd(61232, 0)$$

The second argument is now zero.

Hence:

$$\boxed{\gcd(3918848, 1653264) = 61232}$$

This is exactly the progression shown in the lecture slides.

7.9 Dry-Run Table for the Euclidean Algorithm

Step	a	b	Quotient q	Remainder $a \bmod b$
1	3,918,848	1,653,264	2	612,320
2	1,653,264	612,320	2	428,624
3	612,320	428,624	1	183,696
4	428,624	183,696	2	61,232
5	183,696	61,232	3	0

Once the remainder becomes zero:

```
GCD = 61,232
```

7.10 Why Is Euclid's Algorithm So Fast?

At each step, the numbers become substantially smaller.

The slides state that the values reduce approximately by a factor of two, resulting in roughly logarithmic behavior, and note that even GCDs involving 100-digit numbers require only around hundreds of division steps. Week1-merged

The key point is:

> We do not inspect every possible divisor.

Instead, we repeatedly replace the problem with a much smaller but mathematically equivalent problem.

7.11 Understanding the Logarithm

The term:

$$\log n$$

describes growth where repeated multiplication or division reduces a problem very quickly.

Consider repeatedly dividing 1024 by 2:

```
Plain text
1024
512
256
128
64
32
16
8
4
2
1
```

Only 10 divisions are required.

Why?

Because:

$$2^{10} = 1024$$

Therefore:

$$\log_2 1024 = 10$$

This is why logarithmic algorithms are generally extremely efficient.

7.12 Standard Complexity of Euclid's Algorithm

The slides describe the runtime as approximately:

$$\log(ab)$$

in terms of reduction steps.

A commonly used standard bound for the number of Euclidean division steps is:

$$O(\log(\min(a, b)))$$

The two descriptions express the same broad message needed here:

> The number of Euclidean steps grows very slowly compared with the numerical size of the inputs.

7.13 Interesting Connection Between Fibonacci and GCD

There is a beautiful connection between the two major topics in these slides.

The worst-case inputs for the Euclidean Algorithm are closely related to **consecutive Fibonacci numbers**.

For example:

$$\gcd(21, 13)$$

gives:

$$21 \bmod 13 = 8$$

$$13 \bmod 8 = 5$$

$$8 \bmod 5 = 3$$

$$5 \bmod 3 = 2$$

$$3 \bmod 2 = 1$$

$$2 \bmod 1 = 0$$

Notice:

```
Plain text

21, 13, 8, 5, 3, 2, 1
```

These are Fibonacci numbers in reverse order.

This fact is useful in the formal analysis of the Euclidean Algorithm.

Chapter 8: Comparing Naive and Efficient Algorithms

The Fibonacci and GCD examples teach the same fundamental lesson.

Fibonacci

Naive recursive solution

$$O(\phi^n)$$

approximately exponential.

Improved iterative solution

$$O(n)$$

linear.

GCD

Naive approach

Approximately:

$$O(a+b)$$

under the counting model used in the slides.

Euclidean Algorithm

Roughly logarithmic in the numerical input size:

$$O(\log(\min(a,b)))$$

division steps.

8.1 Why Algorithm Design Matters

Suppose we make a computer twice as fast.

A very inefficient algorithm may still remain unusable.

But changing from an exponential algorithm to a linear algorithm can turn a calculation requiring thousands of years into one that finishes almost immediately.

This is why algorithm design can be more important than simply buying faster hardware.

Chapter 9: Important Terms from the Slides

Term	Meaning
Algorithm	Step-by-step procedure for solving a problem
Input	Data supplied to an algorithm
Output	Result produced by an algorithm
Efficiency	Amount of resources used by an algorithm
Running time	Amount of computational work performed
Space complexity	Amount of memory required
Linear scan	Checking items one after another
Optimization	Finding a better/best solution according to some objective
Recursion	A function calling itself
Base case	Condition that stops recursion
Recurrence relation	Formula defining a value using previous values
Fibonacci number	Number defined using the previous two Fibonacci numbers
Lemma	Proven supporting mathematical statement
Theorem	Important mathematical statement that has been proved
Induction	Mathematical proof technique based on base and inductive cases
Array	Indexed collection of values

Term	Meaning
GCD	Greatest Common Divisor
Divisor	Number that divides another without remainder
$d \mid a$	d divides a
Quotient	Integer result of division
Remainder	Amount left after division
Modulo	Operation that returns the remainder
`a % b`	Programming notation for remainder of a/b
Euclidean Algorithm	Efficient algorithm for computing GCD
Linear time	Running time proportional to n
Exponential time	Running time growing approximately like c^n
Logarithmic time	Running time growing approximately like $\log n$

Chapter 10: What Students Should Learn from Week 1

After studying these slides, a student should be able to explain the difference between a straightforward programming task and a problem for which algorithm design matters. They should understand that two correct algorithms can have radically different performance.

Students should be able to define the Fibonacci sequence:

$$F_0 = 0, \qquad F_1 = 1, \qquad F_n = F_{n-1} + F_{n-2}$$

and calculate small Fibonacci numbers manually.

They should understand why:

```
FibRecurs(n)
```

is inefficient: it repeatedly solves the same subproblems.

They should also be able to dry-run:

```
Plain text

FibList(n)
```

and explain why it requires only linear work.

For GCD, students should know:

$$\gcd(a, b)$$

means the greatest number dividing both a and b.

They should understand the key Euclidean identity:

$$\boxed{\gcd(a, b) = \gcd(b, a \bmod b)}$$

and should be able to repeatedly apply it until the remainder becomes zero.

Finally, students should remember the most important message of the entire lecture:

$$\boxed{\text{A correct algorithm is not necessarily an efficient algorithm.}}$$

and:

$$\boxed{\text{Choosing the right algorithm can completely change the feasibility of a problem.}}$$

The final slide makes essentially this same point: the naive algorithm can be too slow, while finding the right algorithm often depends on discovering an important mathematical property of the problem. Week1-merged